arihantjaingoa@gmail.com
Website

Arihant Jain

LinkedIn

Education

University of California, San Diego — San Diego, CA
B.S. in Computer Engineering, GPA: 3.78/4.0, MGPA: 3.96/4.0 — *Sep 2022 – June 2026*

- **Areas of Interest**: Machine Learning, Robotics & Control
- **Coursework**: Data Structures & Algorithms, Signals & Systems, Embedded Systems, Machine Learning, AI, Deep Learning, Computer Vision, Controls Systems, Autonomous Sensing & Estimation, Planning & Optimal Control

University of California, San Diego — San Diego, CA
M.S. in ML/DS with a focus in Robotic systems and Control — *Sep 2026 – June 2027*

Work Experience

Existential Robotics Lab — San Diego, CA
Undergraduate Researcher — *Nov 2025 – Present*

- Building latent representations and world dynamics models of complex real world interactions for safety filtering.
- Developing sampling-based SDFs on latent dynamics to improve performance in high-DOF and complex scenarios.
- Attempting universal constraint/reward learning for safety and robustness using novel visual embeddings.

Cadel.Ai — Bengaluru, Karnataka
Software Developer Intern — *Jul 2026 – Sep 2026*

- Improved workflow performance and robustness from 69% to over 99.1% using deterministic tools.
- Designed and tested software benchmarking existing workflows, and compared them to existing LLM-based tools.
- Built new tools for workflow optimization, taking advantage of RL-based methods to increase consistency by 1.7x.

Jacob's School of Engineering — San Diego, CA
Research Intern — *Jan 2025 – June 2026*

- Redesigned and improved a multi-modal glove for improved sensor reliability and performance with software.
- Developed a real-time (100Hz) pipeline for patient data collection and assessment using multi-threading.
- Tested a clustering algorithm (HDBSAN) to identify properties in patient data with a test accuracy of over 90%.

Eazy Financial Services — Manama, Bahrain
IT Support Intern — *Jul 2023 – Aug 2023*

- Configured end user hardware and software, including code injection and product deployment onto card machines.
- Spearheaded local server upgrades, replaced RJ-45 cables making custom ones, and made changes to 6 racks.

Projects

Triton Robotics — San Diego, CA
Embedded Lead — *Jan 2025 – June 2026*

- Refactored the embedded codebase, improving performance by 10x and improving readability, along with CI/CD.
- Conducted system analysis and tuning for all our motors, leveraging simulation and autotuning PID.
- Tested and implemented drivers for IMU and communication protocols like SPI, CAN and UART.

Skills

Languages: Cuda, Python, C++, Java, C, MATLAB, SQL, Simulink

Technologies: Git, React, MySQL, REST APIs, Docker, Linux, HTML/CSS, mbed-os, RTOS, ROS2, MoveIt, MotionBenchMaker, OMPL, Gazebo, PyTorch, Tensorflow, MuJuCo, JAX, OpenCV